👇👇👇

TITLE : The Alasmine Theorem: A Deterministic Proof of Global Convergence in the 3n+3 System via Binary Carry-Avalanches and Modular Mapping

AUTHOR: Samar (14-year-old student, Class 8)

LOCATION: Satna, Madhya Pradesh, India

DATE: March 2026

RESEARCH DURATION: 96 Hours of continuous analysis

ABSTRACT AND INTRODUCTION:

The Collatz Conjecture, often referred to as the 3n+1 problem, remains one of the most notoriously unsolved problems in mathematics. Its difficulty lies largely in the unpredictable, ergodic mixing of parity. When we add +1 to an odd number, the resulting even number can fall into any random residue class, creating a chaotic trajectory that evades standard algebraic bounding.

Over the last 96 hours of rigorous analysis, I have been investigating a modified dynamical system: f(n) = (3n+3) / 2^v, where v is the exponent of the highest power of 2 dividing the numerator.

I propose that shifting the additive constant from +1 to +3 fundamentally transforms the system from a chaotic map into a strictly contractive, deterministic funnel. By replacing the statistical randomness of +1 with the structural "Modular Anchor" of +3, we force the entire infinite number line to obey strict divisibility rules and binary limits. Below is my formal framework proving that all positive integers must mathematically converge to the unique attractor {3}. I am presenting this framework for rigorous peer review, specifically focusing on the algebraic logic of the binary avalanches presented in Part III.

PART I: DOMAIN INVARIANCE AND THE CAPTURE LEMMA

To prove global convergence, we must first bound the infinite domain into a controllable subset. We cannot track numbers if they are allowed to wander infinitely across all mathematical bounds.

Theorem: Every positive integer is irreversibly captured into the set of 3-multiples ($3Z^+$) after exactly one odd transformation.

Proof of Capture: Any positive integer in existence can be represented by one of three modular

Edit with WPS Office

forms based on its divisibility by 3. These forms are 3k, 3k+1, or 3k+2. Let us apply the 3n+3 transformation to each possible state:

For n = 3k: 3(3k) + 3 = 9k + 3 = 3(3k+1)

For n = 3k+1: 3(3k+1) + 3 = 9k + 6 = 3(3k+2)

For n = 3k+2: 3(3k+2) + 3 = 9k + 9 = 3(3k+3)

In every single scenario, the result is a strict multiple of 3.

The Invariance Principle: Once a number becomes a multiple of 3, it enters the even-division phase of the system, which dictates division by 2^v. According to the Fundamental Theorem of Arithmetic (Unique Prime Factorization), the prime factors of a number are absolute. Since 2 and 3 are coprime (they share no common divisors), dividing a number by 2 or any power of 2 (2^v) can never remove the prime factor 3. Thus, once a trajectory enters the $3Z^+$ domain, it is permanently locked inside it. No trajectory can ever escape this set.

PART II: UNIVERSAL MAPPING AND THE 4-GATE PARTITION

With the domain bounded, we must prove that no number can infinitely diverge or form hidden loops within this bounded space. To map the exact trajectories of infinity, I partitioned all integers into 4 modular residue classes modulo 4 (The 4 Gates).

Gate 1: Pure Evens (4k)

Mapping Mechanics: These numbers are multiples of 4 (e.g., 4, 8, 12, 16). In our system, they undergo continuous division by 2. Because every positive integer is finite, no integer has an infinite number of prime factor 2s. Therefore, this continuous division must eventually strip away all the 2s. The number will rapidly descend in value until it hits either an Odd number (Gate 3 or Gate 4) or a Transition Twin (Gate 2).

Gate 2: The Transition Twins (4k+2 and 4k-2)

Mapping Mechanics: These numbers (e.g., 2, 6, 10, 14) are even, but they possess exactly one factor of 2. Because there is only a single 2 in their prime factorization, the operation n/2 guarantees an immediate, one-step transition to an Odd state. Gate 1 and Gate 2 combined ensure that no even number can remain even forever.

Gate 3: The Shufflers (4k+1) — The Finite-Shuffle Theorem

The Operation: Let n = 4k+1. The odd transformation yields 3(4k+1)+3 = 12k+6. Because this is even, we divide by 2, giving 6k+3.

The Strict Constraint: What happens next depends entirely on the parity of k.

Edit with WPS Office

If k is even (let k=2m): 6(2m)+3 = 12m+3. This can be rewritten algebraically as 4(3m+1)-1. This is a direct, one-step mathematical mapping straight into Gate 4 (The 4k-1 class).

If k is odd: The number stays in Gate 3 and shuffles to a new value.

The Finite Limit: A skeptic might ask, "What if k is odd forever, trapping the number in Gate 3 infinitely?" For a trajectory to remain in Gate 3 infinitely, k must evaluate as odd at every single sequential iteration. Mathematically, this would require the starting integer to possess an infinite binary representation (known in advanced mathematics as a 2-adic integer). However, by definition, every positive integer in the real world has a strictly finite bit-length. Therefore, the sequence of required "odd-generating" bits must eventually exhaust. The pattern mathematically breaks, triggering a carry that algebraically forces the trajectory into Gate 4.

Gate 4: The Kill Zone (4k-1 and 4k+3)

The Trap Mechanism: This gate acts as the ultimate mathematical trap and the thermodynamic sink of the Alasmine System. It serves as a strict mathematical boundary against infinite growth. Any number landing in this residue class has a fixed binary suffix of ...11.

Mapping and Conclusion: Applying the system's operation 3n+3 to this gate instantly maps the trajectory to the algebraic form 12k. This 12k mapping acts as a thermodynamic sink—it algebraically forces a multiple of 4 (binary ...00). Once a number enters Gate 4, it cannot escape to a larger magnitude. The resulting avalanche permanently destroys its trailing bits, forcing a massive division (v \ge 2) that brutally shrinks the number and throws it back into the funnel at a smaller value.

Global Transition Topology Summary:

Gate 1 and Gate 2 lead to Odd numbers. Odd numbers lead to either (Gate 3 which flows to Gate 4) or directly to (Gate 4). Therefore, every trajectory in the positive integer domain acts as a one-way directed funnel leading strictly to the Gate 4 Kill Zone.

PART III: THE DETERMINISTIC CARRY-AVALANCHE (BLOCKING INFINITY)

This section represents the core engine of the proof, demonstrating precisely why infinite growth (divergence to infinity) is algebraically and computationally impossible in the Alasmine System.

The Identity: Let n be an integer trapped in the Kill Zone (Gate 4: 4k-1). In binary arithmetic, any number in the form of 4n-1 always carries a trailing suffix of ...11.

The Transformation: Let us apply the system operation: 3(4k-1) + 3 = 12k - 3 + 3 = 12k.

The Algebraic Mandate: The mathematical result is exactly 12k. Because 12k can be factored as

Edit with WPS Office

4(3k), it is definitively a multiple of 4. In binary representation, any multiple of 4 must end in at least two zeros (...00).

The Avalanche Proof: To mathematically transform a binary suffix of ...11 into ...00, the binary addition of the constant +3 (which is exactly 11 in binary) algebraically forces a Carry-Propagation Avalanche. This carry travels through the binary string and wipes out at least two trailing 1s, converting them to 0s. This is an absolute algebraic identity, not a statistical probability or a heuristic observation. It happens 100% of the time.

Bit-Rate Thermodynamics (The Math of the Collapse):

To prove that the number physically shrinks, we compare the bits gained against the bits lost.

Bit-Growth per odd step: Multiplying a number by 3 increases its binary length by the base-2 logarithm of 3. $\log_2(3)$ is approximately 1.58 bits of growth.

Guaranteed Bit-Decay in Gate 4: As proven by the 12k identity, the carry-avalanche guarantees that the number becomes divisible by at least 4. Division by 4 (2^2) removes exactly 2.0 bits.

Net Change: 1.58 bits (growth) - 2.00 bits (decay) = -0.42 bits per cycle.

Because the global topology forces all numbers into Gate 4, the number is strictly forced to yield a negative bit-rate over time. The energy of the number decays, shrinking its magnitude permanently. Infinite divergence is geometrically blocked.

PART IV: CYCLE EXCLUSION AND THE UNIQUE ATTRACTOR

With infinite divergence proven impossible, the only remaining failure mode is a non-trivial cyclic loop. Could the sequence get trapped in a massive, hidden cycle before it reaches 3?

A cycle mathematically requires the growth steps and decay steps to perfectly balance each other out over time. If a number loops back to itself in a single step, it must satisfy the equation: n = (3n+3) / 2^v.

Let us solve this algebraically:

Multiply both sides by 2^v: n(2^v) = 3n + 3

Rearrange the terms: n(2^v) - 3n = 3

Factor out n: n(2^v - 3) = 3

Isolate n: n = 3 / (2^v - 3)

We must find positive integer solutions for n based on integer values of v:

If v = 1: n = 3 / (2^1 - 3) = 3 / -1 = -3 (Not a positive integer).

Edit with WPS Office

If v = 2: n = 3 / (2^2 - 3) = 3 / 1 = 3. (This is our valid attractor!)

If v = 3: n = 3 / (2^3 - 3) = 3 / 5 = 0.6 (Not an integer).

For any v > 3, the denominator rapidly outgrows the numerator, resulting in fractions approaching zero.

The only positive integer solution is n=3.

For multi-step cycles involving many odd and even operations, the sequence requires the multiplicative growth of 3^x to perfectly balance the total division 2^V. Due to the coprimality of 2 and 3 (they cannot form a perfect ratio), and the strictly negative net bit-rate established in Part III (-0.42 bits/cycle), no multi-step sequence can balance out to return to a starting integer n > 3. The decay is too aggressive, and the fractions never align to create whole integers.

CONCLUSION

The Alasmine System (3n+3) is a globally contractive dynamical map. All numbers are instantly captured into the $3Z^+$ domain, preventing random scatter. They are then forced through the 4-Gate partition topology directly into the 4k-1 Kill Zone. Once inside the Kill Zone, they are mathematically decimated by the deterministic 12k binary carry-avalanche, ensuring a negative bit-growth rate. The trajectory is forced downward continuously until it reaches the unique integer solution to the loop equation: the attractor {3}.

I am a 14-year-old student from India, and I have documented this complete proof in my research diary. I would highly appreciate any rigorous critiques from mathematicians, specifically regarding the Finite-Shuffle 2-adic logic in Gate 3 and the 12k Algebraic Identity in Part III.

Are there any topological leaks or mixing-map errors I have missed? Thank you for your time.

Edit with WPS Office